

FORM TP 2015313



TEST CODE **02220022**

MAY/JUNE 2015

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

C A R I B B E A N A D V A N C E D P R O F I C I E N C Y E X A M I N A T I O N[®]

E N V I R O N M E N T A L S C I E N C E

A G R I C U L T U R E , E N E R G Y A N D E N V I R O N M E N T A L P O L L U T I O N

UNIT 2 – Paper 02

2 hours 30 minutes

16 JUNE 2015 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each module.
2. Answer ALL questions.
3. Write your answers in the answer booklet provided.
4. You may use a silent, non-programmable, scientific calculator to answer questions.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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MODULE 1

Answer BOTH questions.

1. (a) “Agriculture is an important activity for promoting food security in many Caribbean countries”. Suggest TWO reasons to justify this statement. **[4 marks]**
- (b) The government of a Caribbean country is encouraging farmers to engage in sustainable agricultural practices to reduce some of the environmental impacts of agriculture. Figure 1 shows some of the observed impacts from agriculture in the Caribbean country. Study Figure 1 and answer the questions that follow.

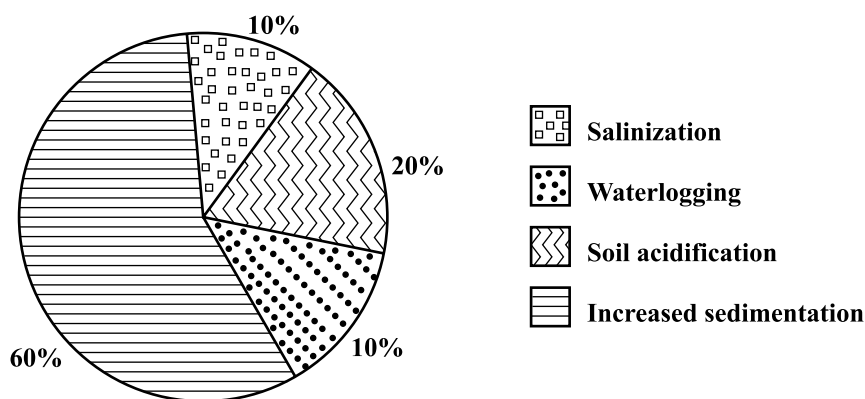


Figure 1. Impact of agriculture on the environment

- (i) State THREE ways by which the practice of sustainable agriculture promotes ecological integrity. **[3 marks]**
- (ii) Using the information presented in Figure 1, make THREE deductions about the impacts of agriculture in the Caribbean country. **[3 marks]**
- (iii) Using the information presented in Figure 1, calculate the total percentage impact on the terrestrial resources of the Caribbean country. Show your working. **[2 marks]**
- (c) State FOUR reasons why the failure to achieve certification of agricultural products to meet international standards may threaten sustainable agriculture in the Caribbean. **[4 marks]**
- (d) Explain how genetic engineering may improve agricultural productivity in the Caribbean. **[4 marks]**

Total 20 marks

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2. (a) Figure 2 shows the amount of soil lost annually from three farms that engage in commercial farming. Different sustainable agricultural farming methods are used on these farms. Study Figure 2 and answer the questions that follow.

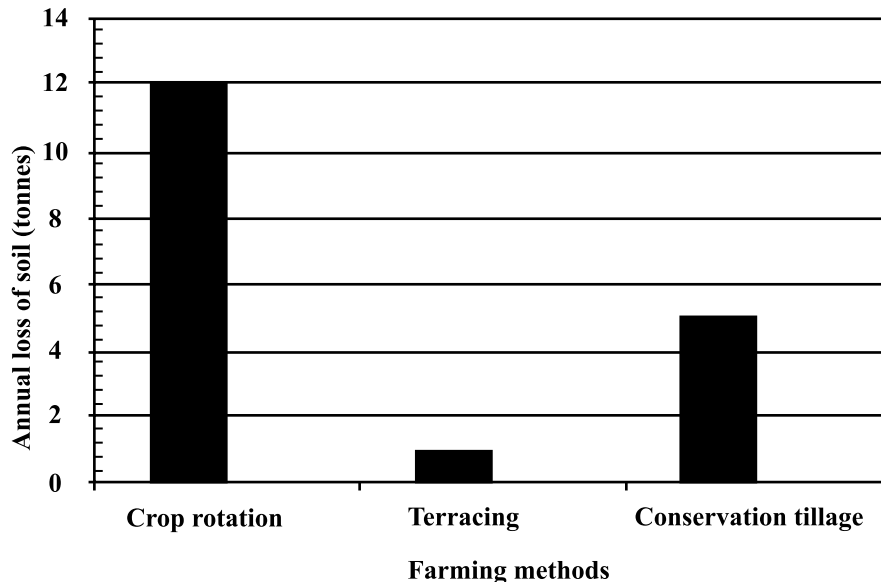


Figure 2. Amount of soil lost annually on farms using different sustainable agricultural methods

- (i) State ONE characteristic of commercial agricultural systems. **[1 mark]**
 - (ii) With reference to Figure 2, state THREE deductions about the annual soil loss from the commercial farms. **[3 marks]**
 - (iii) Calculate the percentage annual soil loss that is attributed to crop rotation. Show your working and give your answer correct to two decimal places. **[4 marks]**
 - (iv) Choose any TWO of the sustainable agricultural methods and for EACH, provide an explanation to account for the amount of soil loss attributed to it. **[6 marks]**
- (b) “In an effort to increase and sustain agricultural productivity, more farmers in the Caribbean should be engaged in the practice of hydroponics”. Justify this statement. Include THREE points in your answer. **[6 marks]**

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 2

Answer **BOTH** questions.

3. (a) (i) Define the terms ‘watt’ and ‘energy conservation’. **[3 marks]**

Table 1 shows data on the cost and number of installations of residential photovoltaic systems in a Caribbean country. Study the table carefully and then answer the questions that follow.

TABLE 1: COST AND NUMBER OF INSTALLATIONS OF RESIDENTIAL PHOTOVOLTAIC SYSTEMS IN A CARIBBEAN COUNTRY

Year	Installed Capacity (MW)	Cost of Installation (\$/W)
2008	150	8.40
2009	400	7.00
2010	600	6.50
2011	1000	6.40
2012	1750	5.20
2013	2200	4.90

- (ii) Summarize in one sentence, the **overall** trend of the data shown in the table. **[3 marks]**
- (iii) If you had to plot a graph showing the relationship between the installed capacity and cost of installation, which set of data should be plotted on the x-axis? **[1 mark]**
- (iv) How many watts are there in a megawatt? **[1 mark]**
- (b) A speaker at a conference states that “the use of solar energy to produce electricity has no impact on the environment”. Evaluate the accuracy of this statement. **[6 marks]**
- (c) The government of the Caribbean country has decided that the energy conservation of domestic energy users needs to be improved. Explain TWO methods the government can employ to improve the energy conservation by domestic energy users. **[6 marks]**

Total 20 marks

GO ON TO THE NEXT PAGE

4. (a) (i) What do the abbreviations 'kWh' and 'TJ' stand for? [2 marks]
- (ii) The recent rapid changes in global oil prices have been described as 'an energy war'. Explain what is meant by this description. [3 marks]

Figure 3 illustrates the energy efficiency of electrical power plants using a variety of energy sources. Carefully study the figure and then answer the questions that follow.

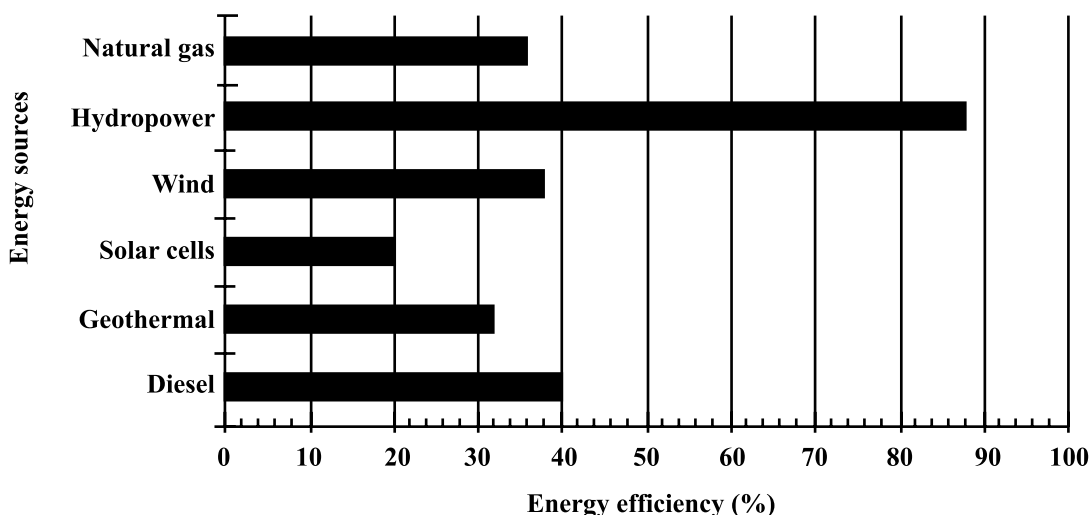


Figure 3. Energy efficiency of electrical power plants using a variety of energy sources

- (b) (i) Arrange the different types of electrical power plants in order of **decreasing** energy efficiency. [1 mark]
- (ii) Use the data for the **FOUR** most energy efficient types of power plants in Figure 3 to construct a suitable table. [4 marks]
- (iii) An electrical plant uses 7.8×10^{17} J of natural gas in one year. Calculate the amount of energy in the natural gas that is wasted during the process. [4 marks]
- (iv) One type of electrical power plant stands out as being the most energy efficient in comparison to the other types. However, it is not commonly used in the Caribbean. Suggest **TWO** reasons why this might be so. [6 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions.

5. Figure 4 illustrates the levels of carbon dioxide found in the earth's atmosphere from 1960 to the present. Study the figure and answer the questions that follow.

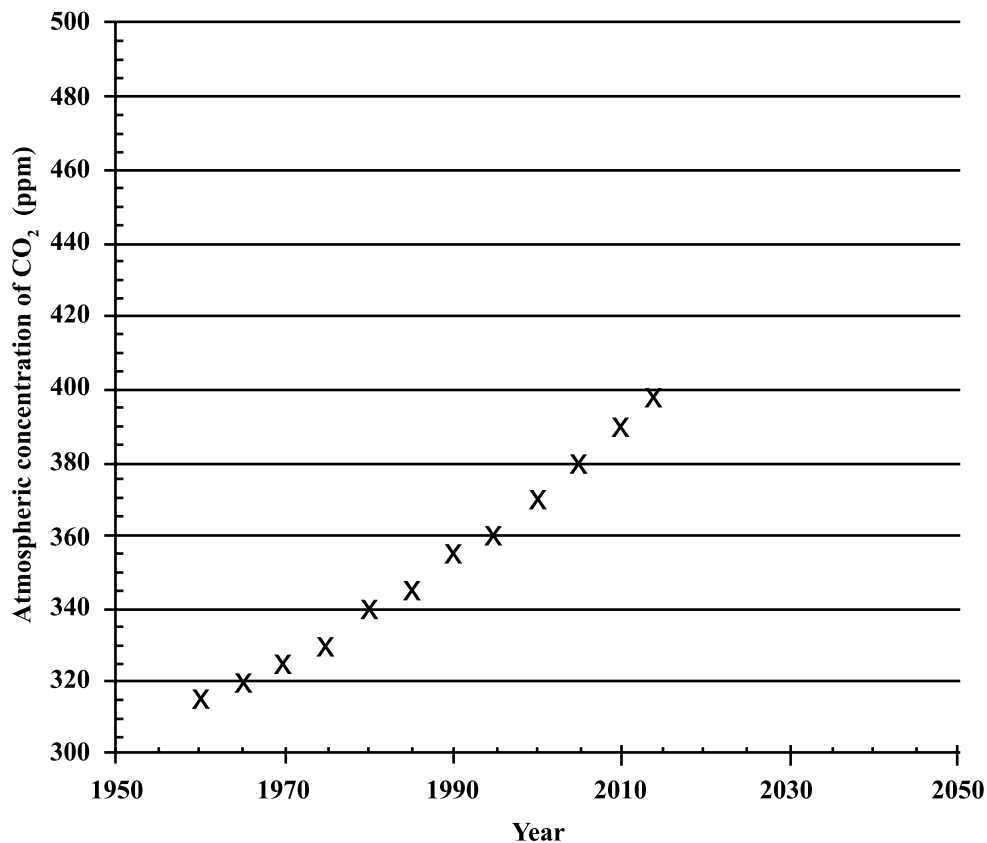


Figure 4. Concentration of atmospheric carbon dioxide as measured at the Mauna Loa Observatory, Hawaii, 1960 to present

- (a) (i) What is the overall rate of increase in carbon dioxide concentration from 1960 to 2010? **[4 marks]**
- (ii) Based on the data in Figure 4, predict the atmospheric concentration of carbon dioxide in 2050. **[2 marks]**

GO ON TO THE NEXT PAGE

- (b) (i) One of the problems with using landfills for waste disposal is that they can ignite, causing landfill fires that are difficult to extinguish. List THREE air pollutants, **other than** carbon dioxide, that could be generated in a landfill fire. **[3 marks]**
- (ii) Air pollution has particularly problematic impacts on receptors in the environment, even those receptors that are a great distance away from the source of the air pollution. What is an environmental receptor AND why is it that air pollution can cause problems for a receptor that is located far from the source of the pollution? **[5 marks]**
- (i) Air pollution is caused not only by landfill fires but also by the use of fossil fuels in vehicles. Explain how TWO technological methods and ONE legislative method can be used to minimize the problem of air pollution from vehicles. **[6 marks]**

Total 20 marks

6. (a) Population lifestyle and consumption patterns are two underlying causes of pollution. In an effort to improve air pollution by changing the energy consumption patterns of its population, a Caribbean country institutes a policy of 100% tax rebate on solar water heaters.
- (i) List THREE underlying causes of pollution **other than** population lifestyle and consumption patterns. **(3 marks)**
- (ii) Explain why changing energy consumption patterns could be expected to improve air pollution. **(4 marks)**
- (iii) Outline a research project to determine the effectiveness of this tax rebate policy at reducing air pollution. In your response, ensure that you clearly identify how the factors you plan to measure will indicate the effectiveness of the policy. **(5 marks)**
- (b) Explain how the United Nations Convention on the Law of the Sea (UNCLOS) functions to protect the environment. **(4 marks)**
- (c) An oil tanker travelling through the Caribbean Sea begins to leak, and a large amount of oil is discharged into the ocean. Explain TWO impacts of this accident on the countries of the Caribbean. **(4 marks)**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.